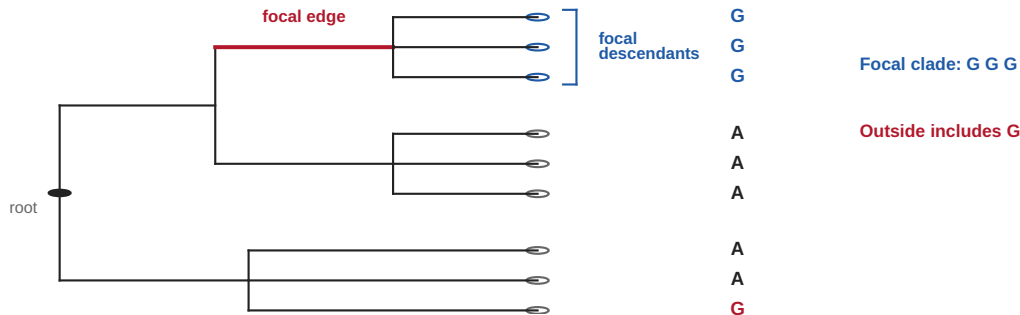


A One branch, one site



B Two questions, different answers

OBSERVED

Clade-exclusive?

focal G G G

outside A A A A A G

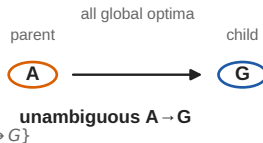
NO not fixed-exclusive
G occurs outside the clade

≠

kept separate

INFERRED

Substitution on the focal edge?



C Retain all optimal focal-edge solutions

Optimal pair set

BRANCHSNV reports

$P^* = \{A \rightarrow G\}$

Unambiguous change

$P^* = \{A \rightarrow G, C \rightarrow G\}$

State ambiguity

$P^* = \{A \rightarrow G, G \rightarrow G\}$

Placement ambiguity

$P^* = \{G \rightarrow G\}$

No change

No arbitrary tie-breaking

All globally optimal focal-edge state pairs are retained.

DESIGN SAFEGUARDS

Explicit root

defines direction

Exact descendants

defines branch

Exact taxon labels

defines correspondence

Complete optimal set

retains uncertainty

Deterministic provenance

supports auditability